# Table-top lab · guide

How Velcro Works

Burrs are for looking, not for faces or mouths; wash hands after handling plants. Pipe cleaner ends can poke: bend them over. Comb strips gently.

### Try 1 · Where did the idea come from?

After a walk outdoors, look closely at your socks. Did any seeds hitch a ride? Look at one with a magnifying glass for hooks.

**Look for:** A burr. In 1941, a Swiss engineer came home from a walk with burdock burrs stuck to his socks and his dog's fur. Under a microscope, each spine ended in a tiny hook. Burrs cling on purpose: animals carry their seeds to new places.

### Try 2 · How do you make a tiny hook?

Bend a pipe cleaner into a small loop. Open one side a little so it becomes a hook. Can your hook catch a loop of yarn?

**Look for:** You start with a loop. The first hooks were cotton and wore out. Nylon worked better: it was woven into loops, heated so the loops kept their shape, then a cutter snipped one side of each loop to leave a hook.

### Try 3 · Why does a peel open so easily?

Press two strips together. Slide them apart sideways, then peel one corner. Which one takes less push? Listen to each.

**Look for:** A sideways pull, called shear, makes every hook hold at once. A peel opens only one thin line of hooks at a time. The rrrip is hooks springing free, one row after another.

### Try 4 · Why do old strips lose their grip?

Find a fuzzy strip on an old shoe or bag. Comb it gently with an old comb or toothbrush, then test the grip again.

**Look for:** Lint and fluff pack into the hooks, so they cannot reach the loops. A fine comb can lift the fuzz out, and the grip comes back.

Print the lab sheet and this guide separately.